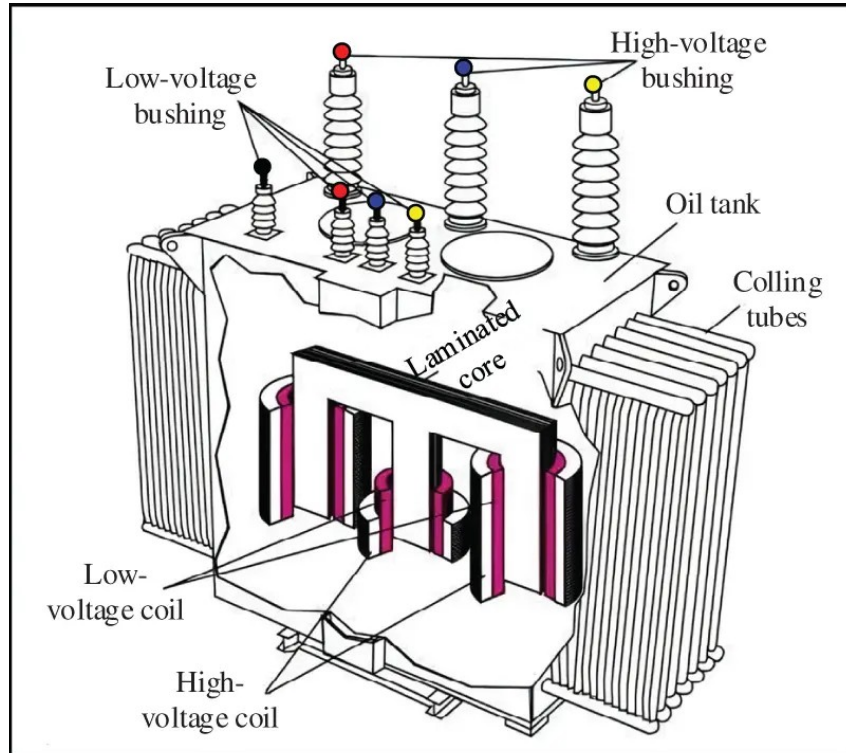


ETTH1

LONG-DURATION FORECASTING OF
TRANSFORMER OIL TEMPERATURES

Johannes Aurich |
28 Aug 2026 |
Final Presentation |

BACKGROUND: OIL IN ELECTRICAL POWER TRANSFORMERS



([Source Link](#))

What is it used for?

- Electrical insulation to prevent arcs and shorts
- Cooling of windings and core
- Limits moisture and air exposure to prevent aging

Why keep track of oil temperature?

- Optimizing asset performance
- Spikes can indicate faults and overload
- Oil can decompose and fail at too high temperatures

DATA FOR LONG-DURATION FORECASTS

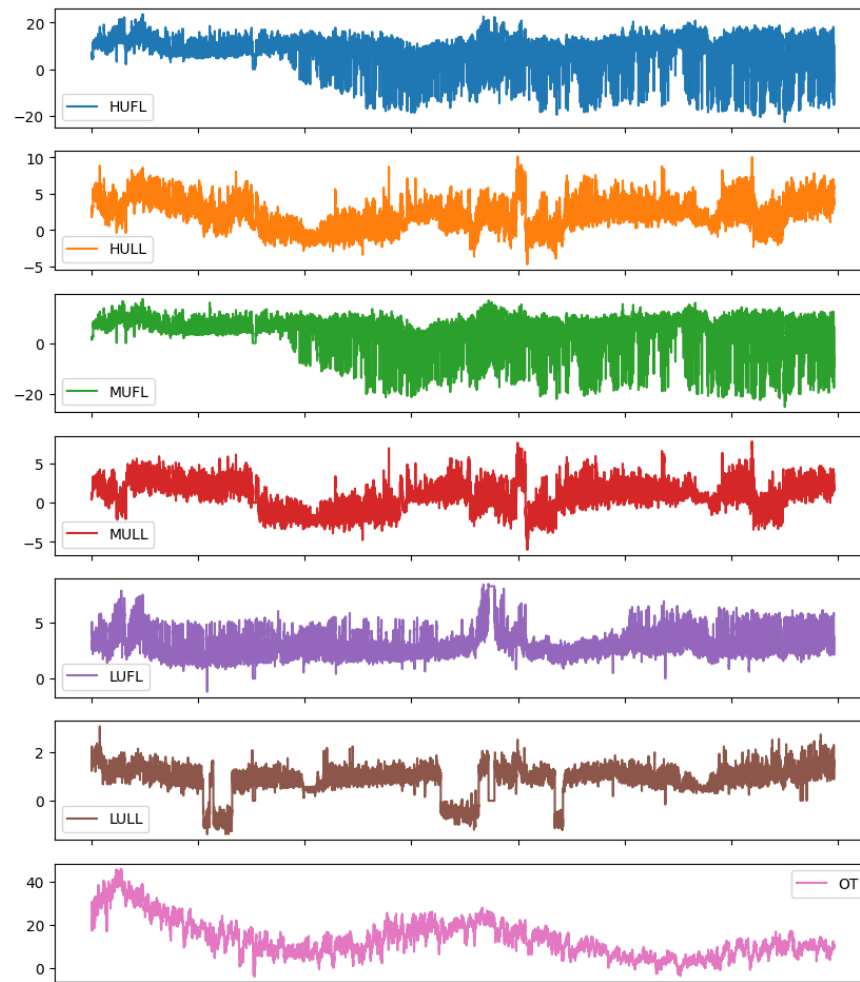
- Electrical substation data from China (pub. 2021)
- <https://github.com/zhouhaoyi/ETDataset>

ETTh1 dataset

- 24 months of data | 1h increments
- 7 channels of transformer measurements
- Typ. lookback & forecasting combinations: 96 – 192 – 336 – 720 h
- **Temporal variation** of frequencies, trends, amplitudes, correlations

Chosen modeling problem

- 96 h forecast, 96 h lookback horizon/context
- Multivariate input to univariate one-shot forecast of OT time series



SPLITTING AND PREPARATION

Using established splits

- 12 training – 4 validation – 4 test (months) | 60/20/20% split
- Discard the remaining 4 months

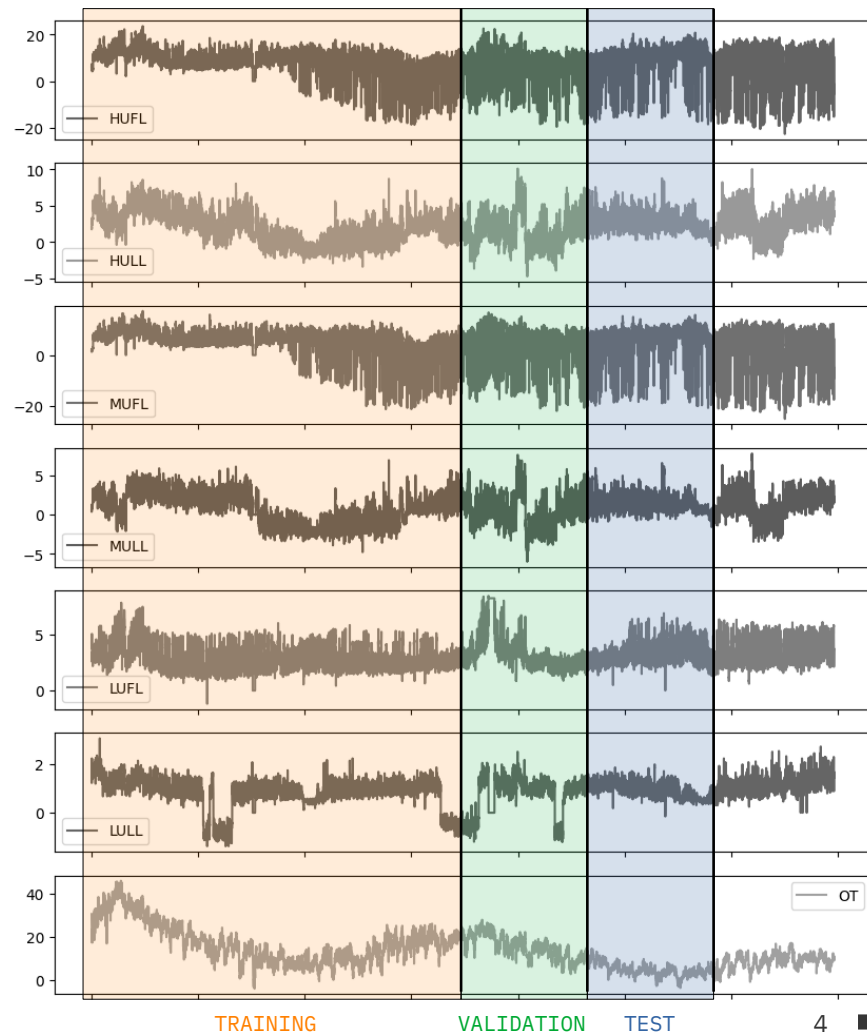
Data set definition

- Rolling-window | stride=1 | 64 samples per batch (133 total)

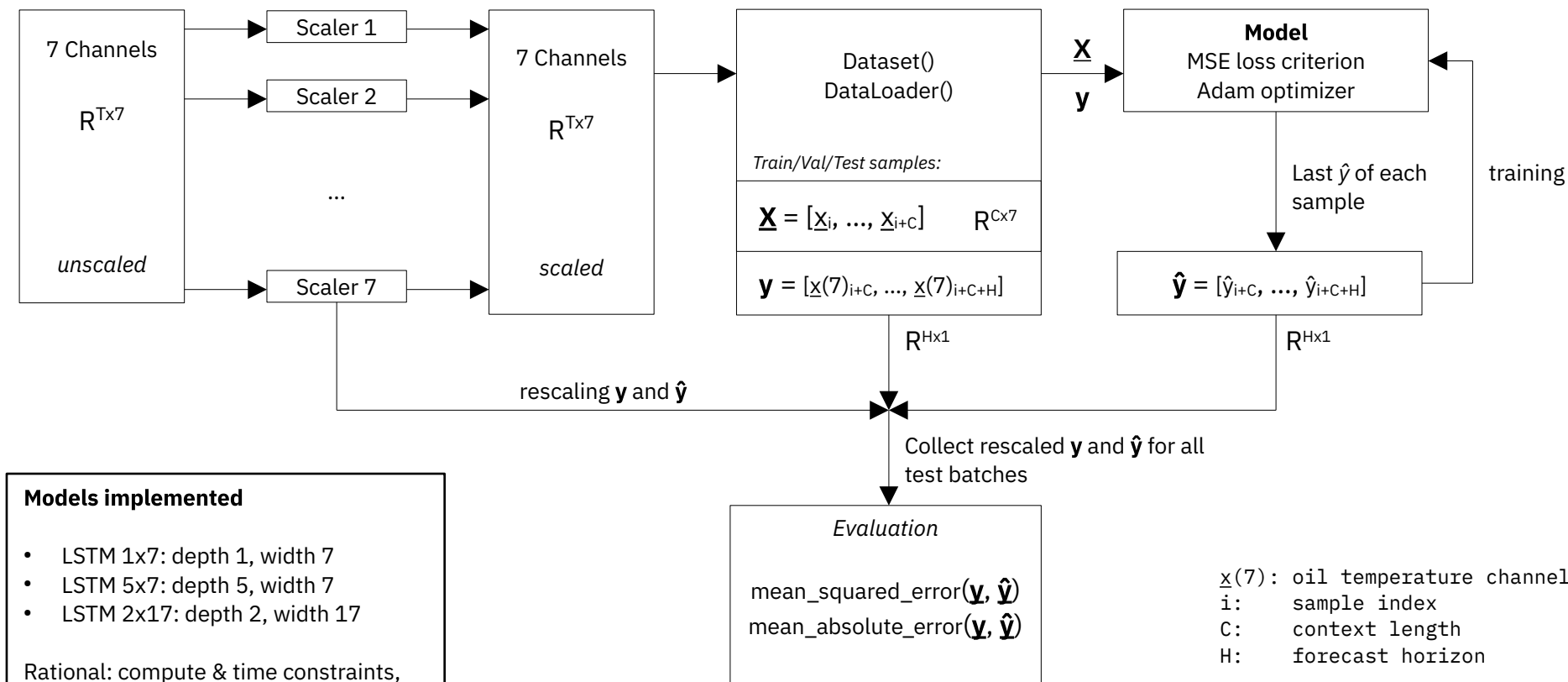
```
training_set = training_split  
validation_set = validation_split + training_split[-context :]  
test_set = test_split + validation_split[-context:]
```

Scaling

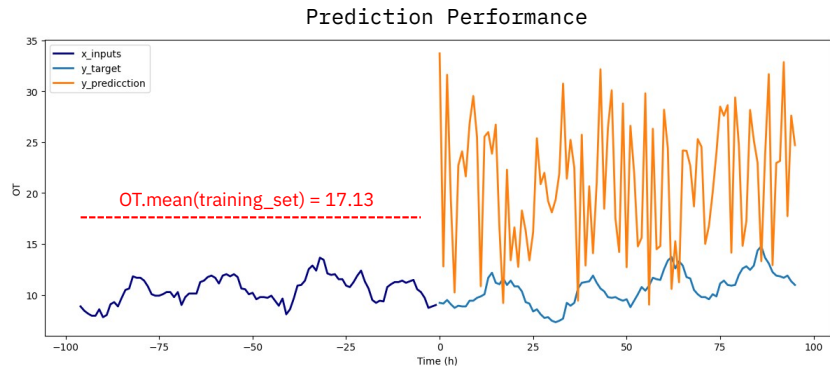
- MinMaxScaler(), feature_range [-1,1]
- Separate scaler per channel, stored for later inverse_transform()
- Per channel: fit on training split, apply to all splits of that channel



MODEL OVERVIEW: MULTI-TO-UNIVARIATE SEQUENCE-TO-SEQUENCE

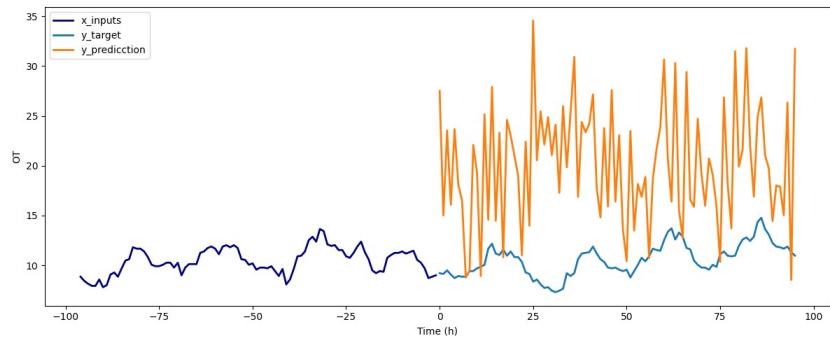


FORECASTING RESULTS



LSTM 1x7 | 1216 parameters

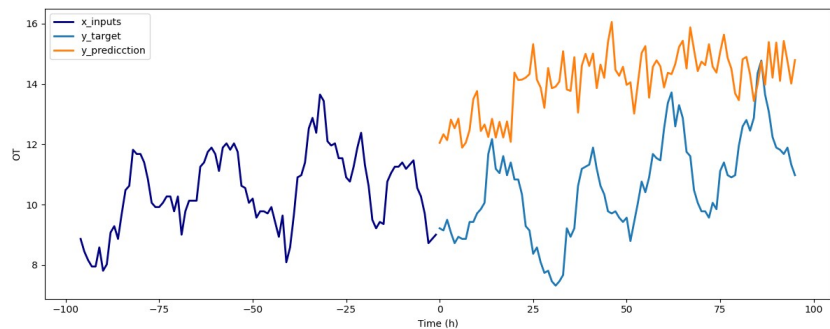
- MSE: 315.50
- MAE: 16.27
- Fails to cover temporal dynamics shifts
→ prob. too few parameters
- Mean is close to mean(training_set)



LSTM 5x7 | 3008 parameters

- MSE: 281.21
- MAE: 15.41

Similar to LSTM 1x7

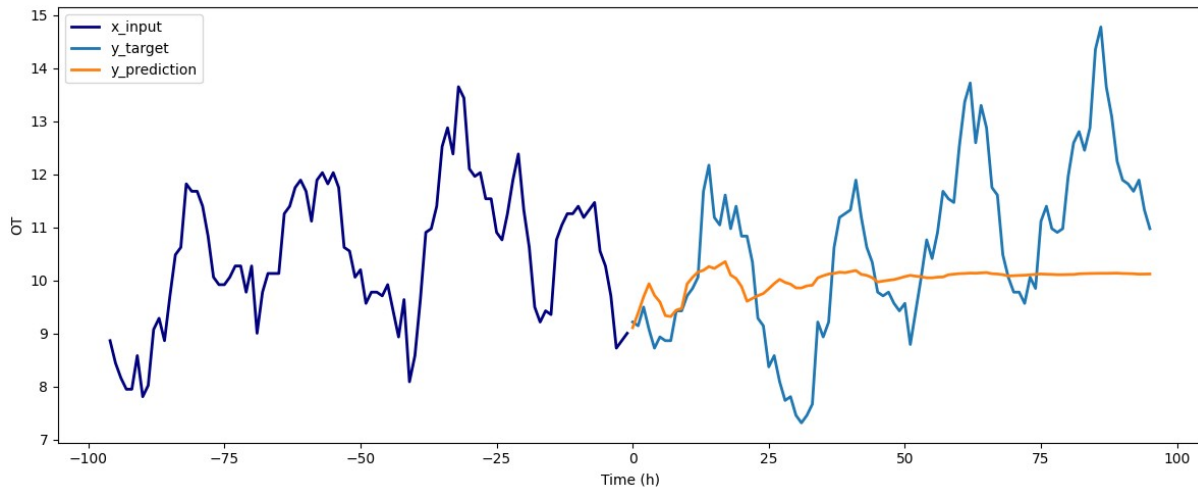


LSTM 2x17 | 5944 parameters

- MSE: 48.29
- MAE: 4.68
- Better in reproducing dynamics
- Still pulled up to mean(training_set)
- TBD if # params or width is the cause
(17 → 96 vs. 7 → 96 output project.)

AN ESTABLISHED STATISTICAL MODEL AS BASELINE: ARIMA

- ARIMA = Auto-Regressive Moving Average
- Based on linear differential equations (= may approximate non-linear processes like ETTh1 at best)
- Most useful for short-term predictions



auto_arima()

- $m = 24$ h seasonality considered
- MSE: 2.62
- MAE: 1.27
- short-term “somewhat” ok
- visible approach of the long-term mean
→ aligned with typical use-cases

- Typical performance of models in the literature: $MSE, MAE \leq O(1)$ covering transformers (very large) to linear baselines (small)
- → 2022 Paper: “[Are Transformers Effective for Time Series Forecasting?](#)”

CONCLUSION & NEXT STEPS

Current model status is highly insufficient

- ARIMA and LSTMs fail (significantly) in providing long-term forecasts, perform worse than literature alternatives
- More analysis of LSTM network architecture and data handling approach required → hyperparameter optimization
 - Longer lookback-context helpful (more information)?
 - Ideal LSTM configuration: depth and width of network

Alternative modelling approaches to explore

- ARIMA captures long-term average + LSTMs may capture short-term dynamics (tbd)
 - Try decomposition approaches: *DLinear* → *DLSTM* or *STL-based* = small & simple but limited by primitive seasonality coverage?
 - Use **State Space Models** (inherent trend and dynamics coverage w/o cutoffs like in *FITS*): S4/Mamba
- Main problem: strong temporal shift in signal dynamics = $f(\text{externalities})$
 - Use **forecast-correction strategies** with recursive loops to for dynamic updates
 - Try to reverse-engineer physical causalities within the system to reduce noise (ambient conditions, heat transfer,...)

THANK YOU! QUESTIONS?